



Parent-Child Emotion Recognition

The role of familiarity in interactive contexts

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**make
history.**



I respectfully acknowledge and pay my respects to the Bedegal people, whose lands we meet on today, and the Kaurna people, the traditional custodians of Tarntanya (Adelaide) where this research was conducted, and where I live and work.

**We rely on facial expressions every day to
read peoples emotions**



**...but how good are we at reading the
expressions of those closest to us?**

Why emotion recognition matters

- Emotion understanding develops through early social interaction
- Parents help shape this development
 - Through emotion socialisation, parents model, label, and respond to emotions, shaping how children learn to express and interpret their feelings (Castro et al., 2015; Eisenberg et al., 1998)
- Children may be better at expressing and interpret emotions with familiar people, though this is largely unexplored

The need for interactive tasks

- Past research relies on posed, static stimuli, often with unfamiliar people.
- But facial expressions are inherently dynamic and context-sensitive (Barrett, 2012)
- We need interactive, ecologically valid methods

How well can parents and children interpret each other's emotional expressions during an interactive task – and does familiarity play an important role?

Building on Previous Findings

Fong et al. (2020)

Children guessed experimenter's expressions in an interactive game

Key Findings

- Recognition accuracy improved with age



Buttle et al. (2020)

Children and parents complete the same game, guessing each other's expressions. Naïve adults guessed children's expressions

Key Findings

- Parents were more accurate than naïve adults. Potential effect of familiarity.
- No effect of child age.



The Current Study

Further investigating this familiarity effect

The Current Study



- 45 parent-child dyads, children aged 4 to 12
- Recruited and tested at the South Australian Museum (N = 32) and in the Develop Well research lab (N = 13)
- Participants completed an object guessing game with each other, then with video clips of unfamiliar parents and children
- Expressions were video recorded, and we measured guess accuracy in each trial
- Video recordings were shown to naive adults



What we predicted

- Parents would perform better in the expression recognition task with their own child, compared to other unknown children
 - Parents average task performance would improve with child age
- Children would perform better in the expression recognition task with their own parents, compared to other unknown parents
- Familiar parents will perform better at recognising expressions of their own child compared to naïve parents, and both will perform better than non-parents

Method



8 boxes



2 of each object

Method

Parents and children take turns being the **Expressor** and the **Guesser**

The expresser is handed a box, and is instructed to open it, look inside, and then make a face to show how they would feel if they saw the object that is inside

The guesser sits in front of examples of the objects, and points to which one they think the expressor saw.

Method

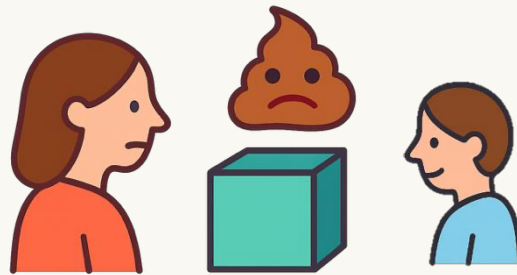
We measured parents and children's accuracy in each trial.

- Did they correctly guess the **object presented**?

Compared this performance across guesser and across task type.

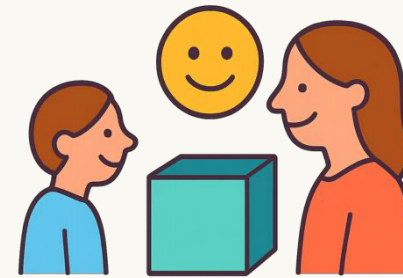
Familiar Person Box Task

1



Round 1

Parent is the expresser
Child is the guesser



Round 2

Child is the expresser
Parent is the guesser

Unfamiliar Person Box Task

2



Parent watches videos of
other age-matched children
expressing



Child watches videos of
other parents expressing
(to age-matched kids)

Object Emotion Labelling

3

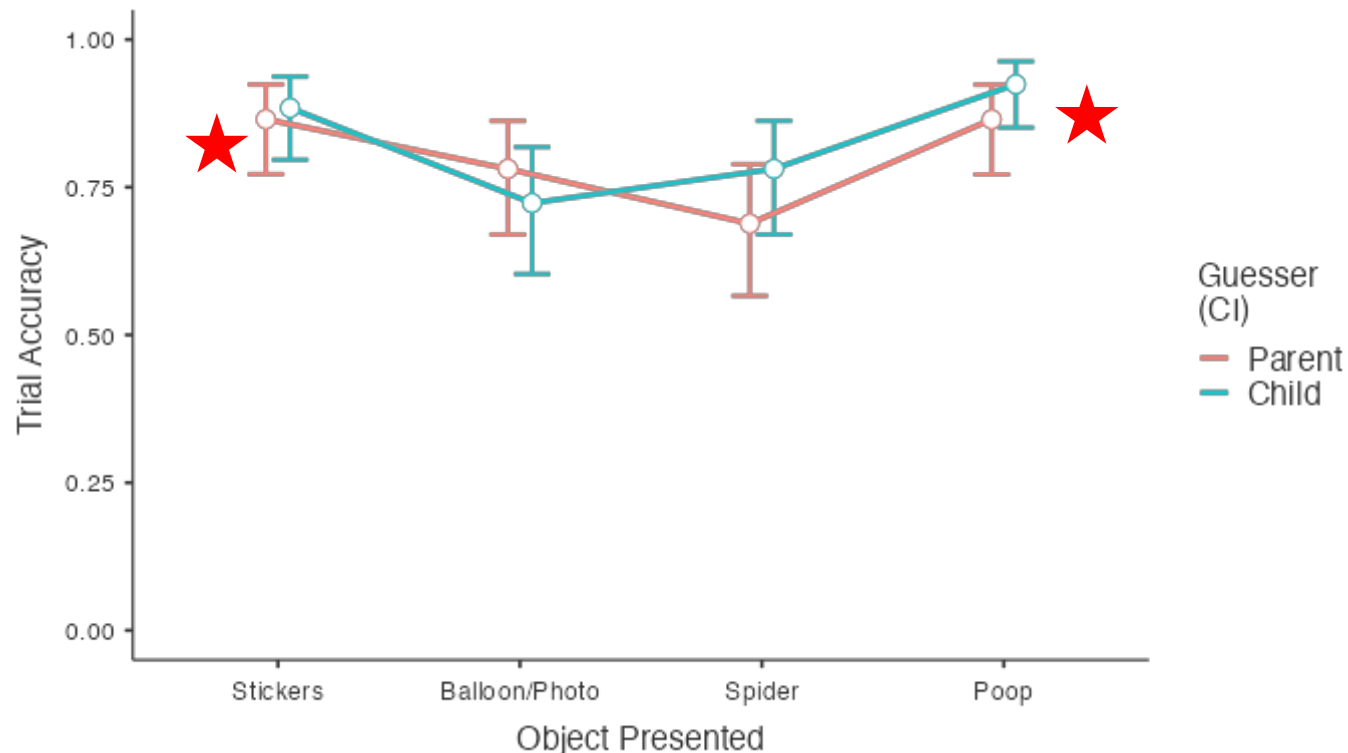


Child labels the emotion they
would associate with each object
(free-choice labelling)

Results – familiar trial

Parents and children perform similarly with each other

Parent and child trial accuracy across object presented

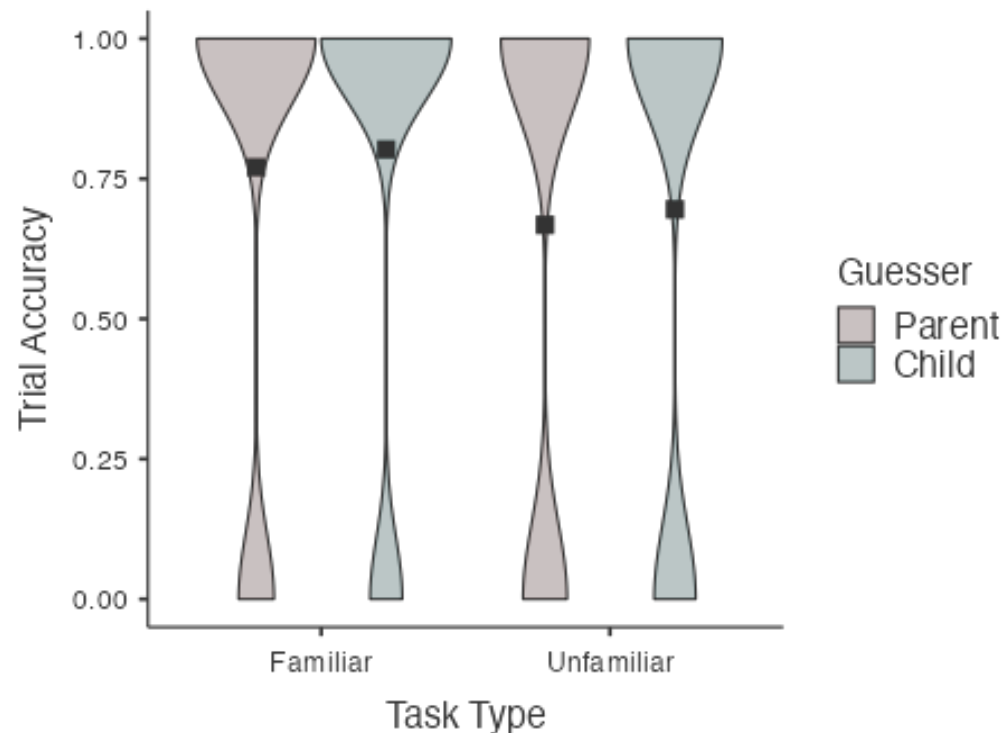


- No significant difference between **parents and children** in their **accuracy** with each other
- **Object presented** influenced recognition
- **Child age** was a weak but significant predictor of performance ($p = 0.049$)

Results – unfamiliar *and* familiar trial

Familiarity improves accuracy

Mean trial accuracy of parents and children across the familiar and unfamiliar task



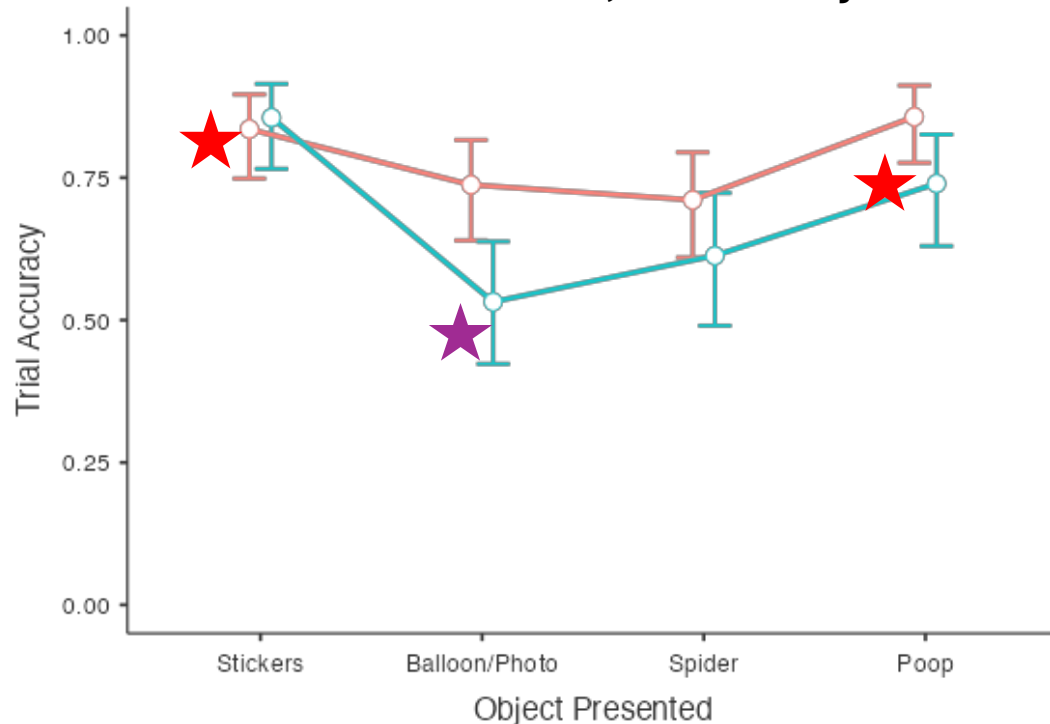
→ **Parents *and* children** were more accurate with each other, than with unfamiliar parents and children

Results – unfamiliar *and* familiar trial

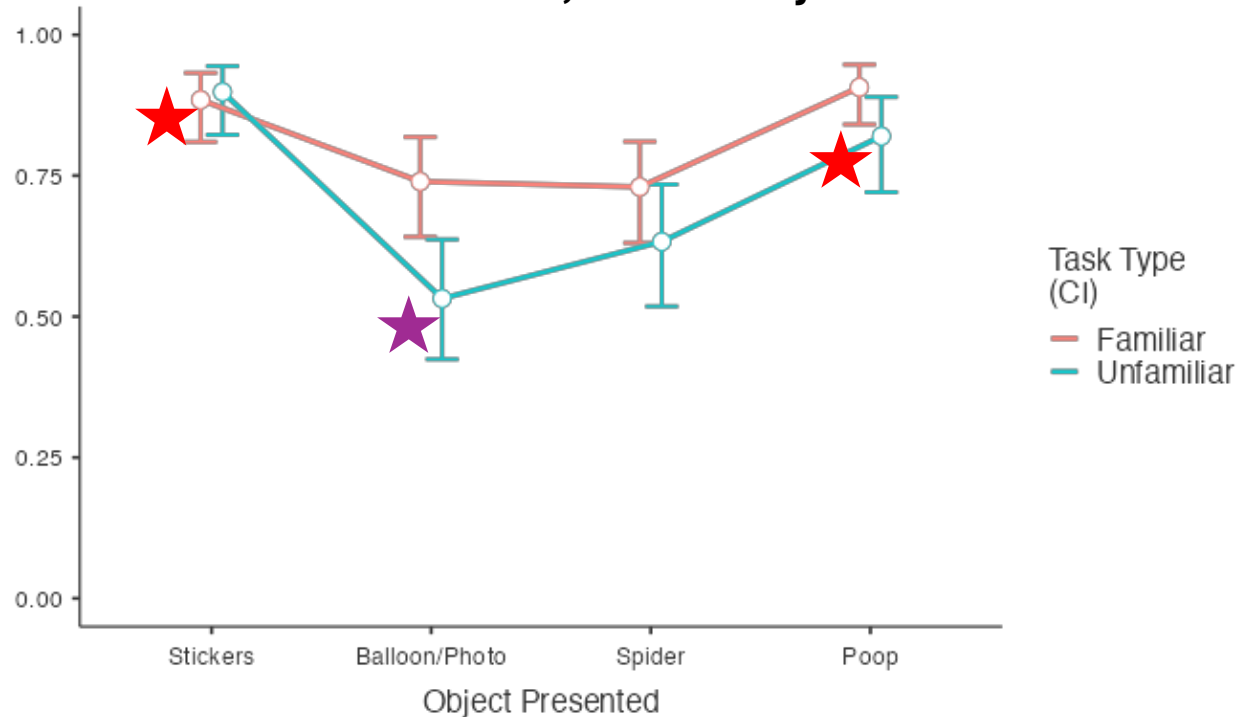
Familiarity improves accuracy

→ **Object type** influenced recognition

Parents performance in familiar and unfamiliar tasks, across object



Children's performance in familiar and unfamiliar tasks, across object

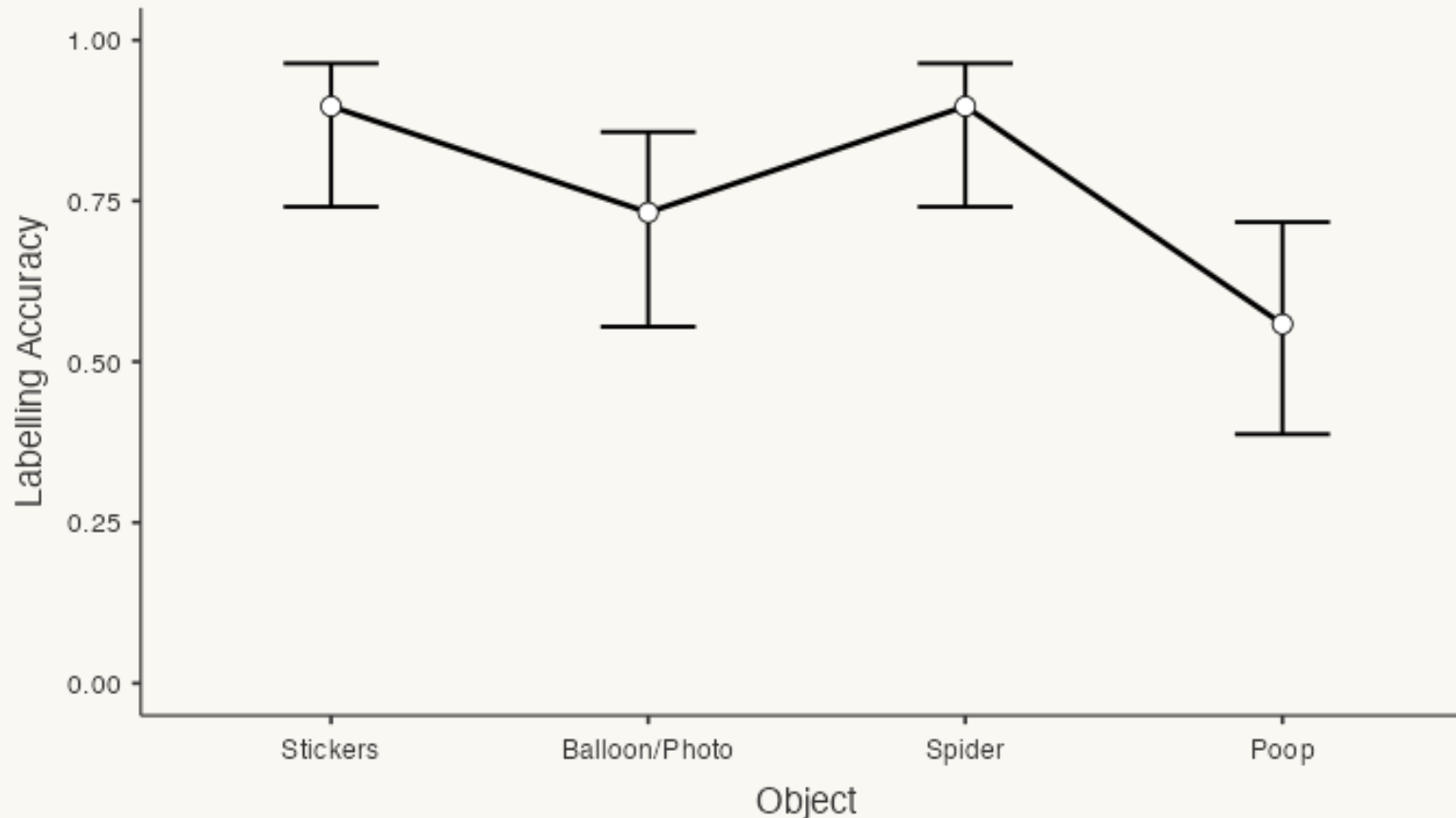


Results

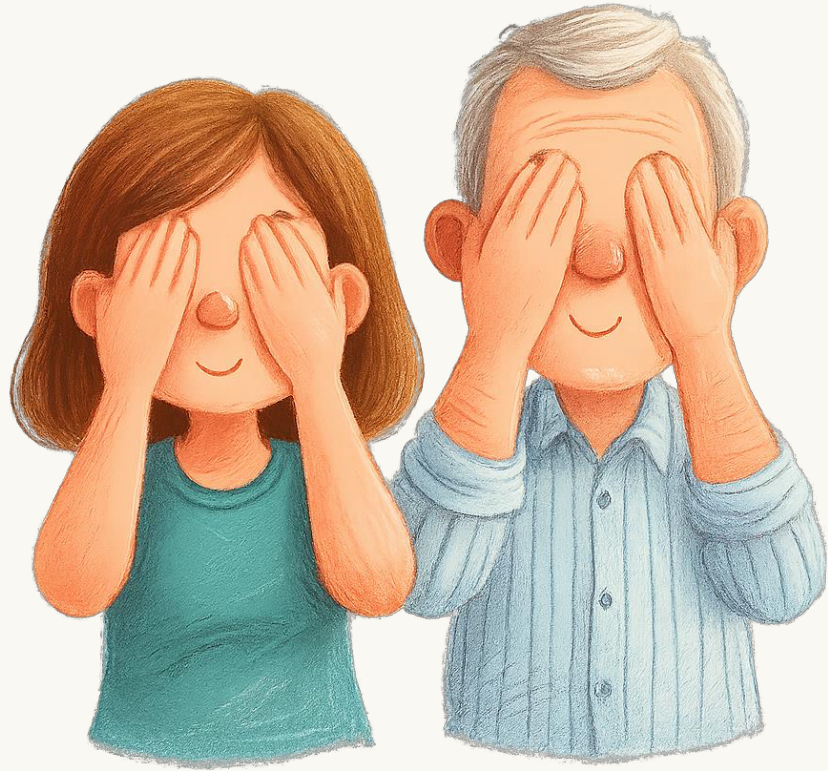
Free-labelling of emotions

Children's free labels were compared to a word bank of accepted labels for each target emotion (Happy, Sad, Fear, Disgust)

Children's emotion labelling accuracy across object



- **Object type** influenced labelling accuracy
- Children were **less accurate** labelling *poop* compared to *stickers*
- No significant differences between *stickers* and *photo* or *spider*



Naive adult raters

To explore the familiarity effect further, video recordings from Study 2 were presented to naive adults

We recruited 69 parents (of similarly aged children) and 68 non-parents

Naive Adult Box Task

Parents



Recruited parents with
children

Allocated to an age group based
on own child/ren's age, then
watched 8 randomised videos of
children expressing

Non-Parents



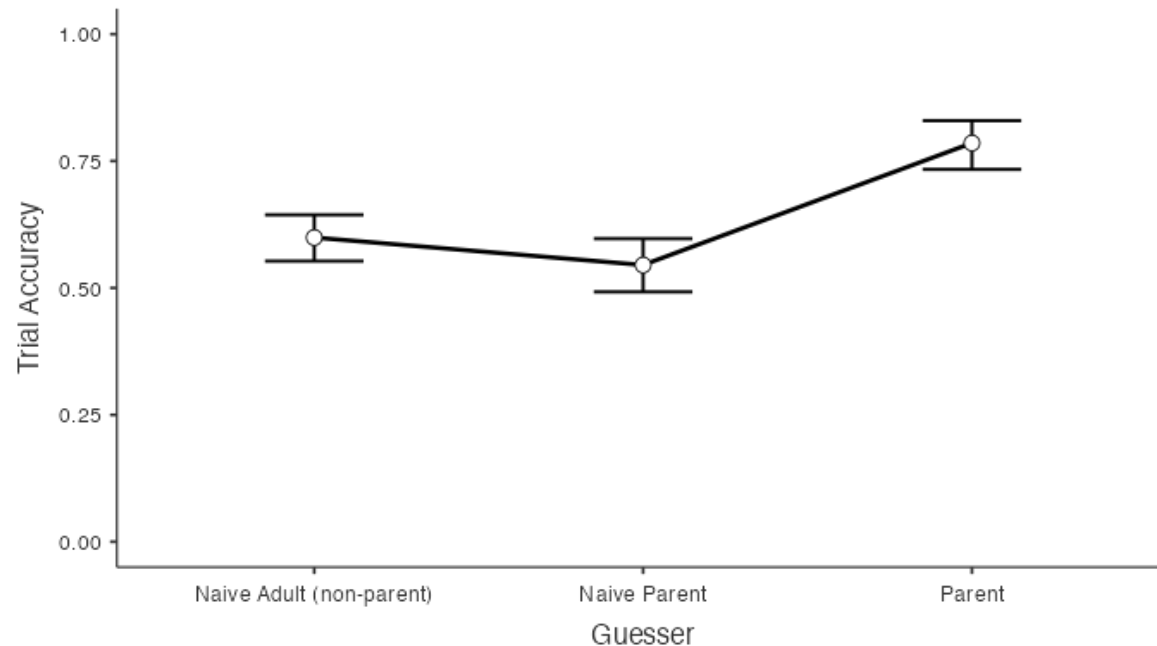
Recruited adults who did not
have children of their own

Randomly allocated to an age
group and watched 8 randomised
videos of children expressing

Results

Parent-specific advantage

Mean trial accuracy across non-parents, naive parents, and familiar parents

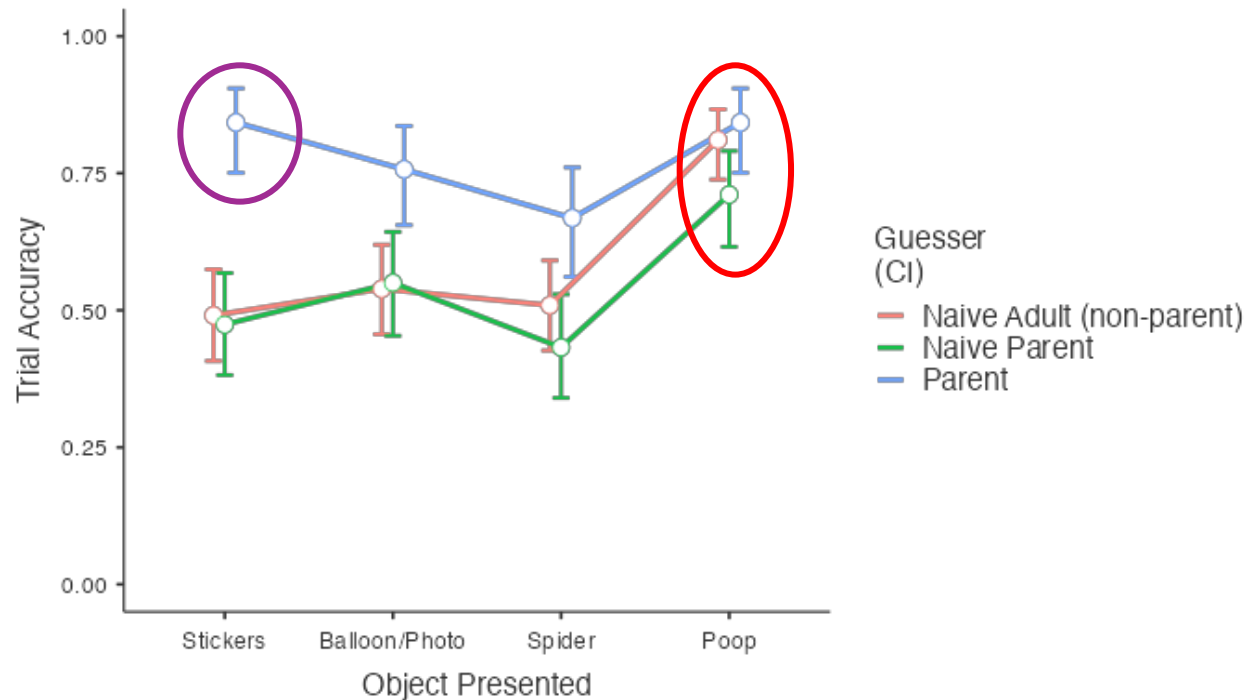


- **Familiar parents** were significantly more accurate than both **naive parents** *and* **non-parents**
- No significant difference between **naïve parents** and **non-parents**

Results

Parent-specific advantage ...for happy expressions!

Mean trial accuracy for naive adults and parents, across object presented



- **Familiar parents** were significantly more accurate than both **naive parents** *and* **non-parents** on sticker trials
- Previous research has highlighted that children struggle to purposefully express happy in a way that is recognisable to adults (Fong et al., 2020)
- Everyone did quite well with disgust!

Discussion & Implications

Familiarity enhances emotion recognition

- Parents and children performed better with each other.

Parents display an advantage in recognising their *own* child's emotions

- Familiar parents outperformed ALL naive adults (including other parents)

Familiarity helps parents to interpret their child's happy expressions

- Interesting given that children typically don't express happy clearly

No strong age effects

- Accuracy didn't consistently improve with age – relationship may be more important than developmental stage?



Future Directions

Of course, parents aren't the only 'familiar' adult figures in children's lives!

In the future, this study design could be extended to explore how teachers, grandparents, or other extended family all interpret children's emotions.

What's next?





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